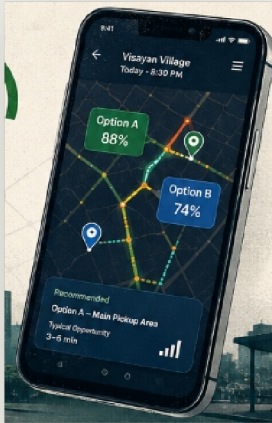
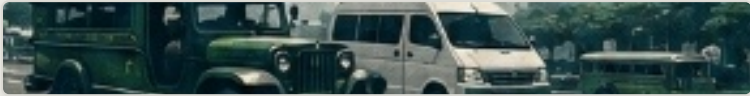


ROUTE SENSE

Know **where** rides are.
Know **when** they're available.



- SMARTER**
Find the best route
- REAL-TIME**
Up-to-date information
- CONNECTED**
Know your network
- PRIVACY**
Your data is secure

1 THE PROBLEM

Knowing a transport route does not mean knowing whether a ride is actually available.

Informal public transportation can change depending on the time, location, demand, traffic, weather, and day of the week. Commuters may already know where vehicles usually pass, but they still face questions such as:

- ? "Can I actually get a ride here right now?"
- ? "Where is the better place to wait?"
- ? "How reliable is transportation at this hour?"
- ? "Will I still have mobile signal during the trip?"

Existing static route maps cannot fully answer these changing, real-world conditions.

2 BENEFICIARIES

PRIMARY USERS

- Students and daily commuters
- Workers and late-night commuters
- Delivery and gig-economy workers

SECONDARY BENEFICIARIES

- Schools and universities
- Transport cooperatives and associations
- Local government and transport planners
- Researchers
- Telecommunications and mobility planners

3 THE SOLUTION

RouteSense is an Android-based app that uses sensors already available inside smartphones to understand how informal transportation works.

- StopSense**
Learns patterns related to passenger movement, probable pickup areas, transport activity, and service availability.

Instead of simply showing commuter routes, RouteSense provides estimates:

- Where rides are likely available**
- How reliable the estimate is**

4 HOW IT WORKS

- Opt In**
The commuter voluntarily enables passive mobility sensing.
- Sense Movement**
The phone detects activity changes such as walking, waiting, and travelling in a vehicle.
- Collect Context**
GPS/GNSS, accelerometer, gyroscope, time, and cellular signal measurements provide mobility context.
- Analyze On-Device**
RouteSense estimates probable boarding areas, travel corridors, movement patterns, and connectivity conditions.
- Protect Privacy**
Detailed raw movement data is minimized and converted into anonymous or aggregated observations.
- Build Community Intelligence**
Observations from many trips reveal service patterns, pickup activity, reliability, and connectivity.
- Recommend**
RouteSense suggests transportation options based on current and historical conditions.

5 KEY FEATURES

- 88% Service Probability**
Estimates how likely a commuter is to find transportation at a particular location and time.
- Smart Pickup Areas**
Identifies locations where successful transport activity frequently occurs.
- Opportunity Window**
Shows a realistic waiting range such as **Typical ride opportunity: 4-7 minutes** instead of pretending to know an exact vehicle arrival time.
- Time-Based Mobility**
Learns how transport availability changes between morning, afternoon, evening, weekdays, and weekends.
- Confidence & Freshness**
Every recommendation shows how much recent evidence supports it.

High Confidence • 31 observations at similar times • Last observation: 8 minutes ago
- Connectivity-Aware Routes**
Compares transportation choices based on both mobility and cellular connectivity.
- Route Pattern Discovery**
Detects recurring and emerging transport corridors from aggregated passenger movement.
- Offline Cached Insights**
Previously learned mobility information remains accessible during weak connectivity.
- No Driver App Required**
RouteSense learns from participating passenger smartphones rather than requiring every vehicle to join the platform.

No dedicated GPS tracker or driver app

- Smartphone Sensor Fusion:** Combines multiple sensors to estimate location more accurately than GPS alone.
- Activity Recognition:** Identifies transitions between activities like Walking, Waiting, and Driving.
- GNSS / Geospatial Analytics:** Analyzes movement patterns across large areas.
- Cellular Signal Intelligence:** Measures network strength to infer passenger journeys.
- Context-Aware Computing:** Considers local conditions when interpreting mobility.
- On-Device Inference:** Processes sensitive data locally to minimize results.
- Geospatial Clustering:** Finds recurring pickup and drop-off hotspots.
- Predictive Analytics:** Estimates service probability based on historical observations.
- Privacy-Preserving Data Aggregation:** Builds insights without publishing individual travel histories.

PHONE HARDWARE USED

- GPS / GNSS
- Accelerometer
- Gyroscope

7 TECH STACK

- Mobile App: React Native + TypeScript
- Android Sensor Layer: Kotlin
- Local Storage: SQLite
- Maps: MapLibre + OSM
- Geospatial Backend: Supabase + PostgreSQL
- Geospatial Processing: PostGIS + H3
- Data Analytics: Python + pandas + GeoPandas

9 PROJECT TEAM

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- Anjelo Vidal

CCE 106 — Application Development & Emerging Technologies | Student Researchers

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